Qihan (Blake) Wang

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Education

University of Illinois Urbana-Champaign Urbana-Champaign, IL
B.S. in Electrical Engineering *Expected May 2030*

- **Relevant Coursework:** Differential Equations; Calculus III; Linear Algebra with Computational Applications; Discrete Structures; Introduction to Electronics

Research Experience

Research Intern Summer 2025
University of Houston, Department of Computer Science

- Developed Python scripts to evaluate large language model reasoning accuracy and cross-domain knowledge transfer across biology, mathematics, and law.
- Contributed to research investigating LLM reasoning performance and knowledge transfer under Prof. Shi and Prof. Alsagheer.

EEG Motor-Imagery Classification and Signal Processing 2026
Independent Research Project

- Built a Python/MNE EEG preprocessing pipeline implementing bandpass filtering, epoch segmentation, and frequency-domain analysis.
- Implemented time-, frequency-, and spatial-domain EEG features, including band power, Hjorth parameters, wavelet energy, CSP, and FBCSP; evaluated SVM, random forest, logistic regression, and shrinkage LDA classifiers.

Projects

Chronodox Strategic Simulation Platform | *React, TypeScript, Zustand, Vite, Node.js* 2026

- Developed a browser-based multiplayer strategy simulation with a rule-based engine, procedural map generation and spatial algorithms, and an in-browser terminal interface that exposes game actions, diagnostics, and training workflows
- Implemented REST-based client–server synchronization, built a browser-based AI self-play pipelines and trained a hybrid heuristic/candidate-based gameplay agent using 200,000+ self-play samples.

Void Navigator | *Java, Swing, Java2D, CheerpJ, Xcode* 2025

- Developed a Java arcade game featuring Java2D rendering, physics-based movement, procedural environments, and persistent game state.
- Implemented AI-controlled enemies and Boids-inspired swarm behavior.
- Packaged the game as a native macOS application using Xcode and deployed a browser-playable version on GitHub Pages using CheerpJ; currently optimizing runtime performance and Java-to-JavaScript conversion overhead.

Competitions & Leadership

CyberPatriot | *Platinum Semifinalist, 5th Place NJROTC Semifinals* 2025

- Hardened Linux systems in virtualized environments using Bash, system administration, and cybersecurity tools.

National Program Director, Steel City Codes 2025–2026

- Supported computer science education through curriculum development, outreach, and program coordination; created digital outreach content generating approximately **50,000 views**.

Technical Skills

Languages: Python, C++, Java, Bash, TypeScript, JavaScript, SQL
Scientific Computing: NumPy, MNE, signal processing, machine-learning evaluation
Systems & Tools: Linux, Git, virtual machines, SSH, terminal tools
Web Development: React, Node.js, HTML, CSS
Spoken Languages: English, Mandarin